CIS 371 Web Application Programming

Syllabus

GRAND VALLEY STATE UNIVERSITY

Lecturer: Dr. Haoyu Li

About this course

In this course, we dive deep into how websites and web apps are built and run. We'll look at the common methods and tools used in web development and study the different programming and markup languages that make websites work. We'll also cover important topics like how to connect websites to databases, making sure websites meet standard guidelines, and how to keep websites safe from security threats.

Course Objectives:

After completing this course, students should be able to:

- Apply foundational knowledge of HTML to structure web content effectively.
- Create stylesheets to control the visual appearance of web pages.
- Craft modern front-end web applications using JavaScript and TypeScript.
- Write server-side scripts to support the functionalities of a web application.
- Utilize Ajax to fetch data from the server and dynamically update content without reloading the entire page.
- Employ web APIs effectively to send and receive data between the client and server.
- Analyze large-scale problems and construct comprehensive websites as solutions.

Contact Information

Instructor:	Dr. Haoyu Li
E-mail:	lihao@gvsu.edu
Office:	MAK D-2-220
Office Hours:	Tuesday, Wednesday, and Thursday, 1:00 – 2:00 p.m. in-person and Zoom Other meeting times are available by appointment via Zoom.
Classroom:	Online Asynchronous
Class time:	Online Asynchronous
Final Present.:	Due end of the examination week, December 19, 2026.

Course Logistics & Prerequisites

- No assigned textbook
  - most materials about web development is available online
- Check **Blackboard** on a regular basis for
  - lecture videos,
  - announcements,
  - assignments
- Check **Course Website** regularly

Grading

Course Component	Overall Weight
Attendence Quizzes	5%
Quizzes	20%
Assignments	35%
Projects	40%
Total	**100%**

The instructor reserves the right to make minor adjustments to the point distribution.

Grading

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Assignments, Due Dates

1) **Due dates**: All assignments will be due at 11:59 pm Michigan time on the due date.
2) **Late Policy**: Work submitted after the due date will incur a 10% late penalty per day, with a maximum penalty of 50% (Weekends are not counted). **No assignment will be accepted after the same weekday the following week. All assignments, graded discussions, quizzes, exams, etc., are submitted electronically. No assignments are accepted via email, printed, or any other method.**

   If you are struggling with meeting deadlines while working remotely, please contact me as soon as possible!

Participation

a. This course uses brief weekly participation quizzes to verify engagement with course materials. Each quiz must be completed by its posted deadline. Students who complete at least 80% of the participation quizzes will receive full credit for this portion of the course grade.
b. There will be no make-up participation quizzes unless prior arrangements are made or a documented excuse (e.g., illness, emergency) is provided.
c. University policy states: "In case of excessive absences, the instructor may refuse to grant credit for the course." GVSU Attendance Policy.

Academic Honesty

- ☞ Document Collaborations: Clearly note any collaborations on individual assignments.
- ☞ No Code Sharing: Direct electronic transfer of code between students is not allowed.
- ☞ Cite Internet Sources: If you use code from the internet, provide an active link and ensure it doesn't constitute the entire solution.
- ☞ Engage Online Respectfully: Participate in forums for discussion, not for sharing solutions or soliciting complete answers.
- ☞ Discuss Conceptually: Talk about problems using non-technical, conceptual language rather than sharing specific code.
- ☞ Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.

AI Policy

- Using AI help for the assignments are allowed, not not encouraged.
- Always quote the code and answers from AI.
    - Quote it like this:
    - // --- start AI code ---
    - // --- end AI code ---
- Using AI generated code without quoting is considered academic misconduct and will result in 0 for the assignment
- Mistakes in the code from AI help will results in more point deduction in the assignment

Prerequisites

- Formal Prerequisites: (CIS 162 or CIS 260) and (CIS 333 or CIS 353)
- Fluent in any object-oriented(OO) Languages
  - You should be able to solve most problems at codingbat.com in a couple of minutes. If you struggle in solving these problems, then ***you are not ready*** to take this course
- Self Learner
  - Skilled in high-level programming **concepts**, and able to teach yourself the basics of other C-like languages (Java|Type)Script
- Good understanding of OO techniques: inheritance, methods, interface, ...

Programming Tools

Debugger (in browser)

Recommended

Programming Tools

Visual Studio

Visual Studio Code

Academic Resources

- **The writing center:** The Fred Meijer Center for Writing, with locations at the Allendale and Pew/Downtown Grand Rapids campuses, is available to assist you with writing for any of your classes. For more information about these services and locations, please visit their website: http://www.gvsu.edu/wc/
- **Speech lab:** The Grand Valley Speech Lab is a peer-to-peer communication center that helps students with all elements of oral presentations. For more information about this service, please visit their website: https://www.gvsu.edu/speechlab/

Academic Resources

- **Research consultants:** The Center for Scholarly and Creative Excellence (CSCE) promotes a culture of active, engaged, ethical scholarship. It supports innovative faculty and student research and collaborative partnerships in the broader community. For more information, please visit their website: https://www.gvsu.edu/csce/
- **Library:** GVSU's library offers a vast collection of online resources available for students. Visit their website for more details: https://www.gvsu.edu/library/
- **Disability support resources:** If any student in this class has special needs because of a disability, please contact Disability Support Resources at http://www.gvsu.edu/dsr/ (DSR) at 616-331-2490.

Tentative Course Content

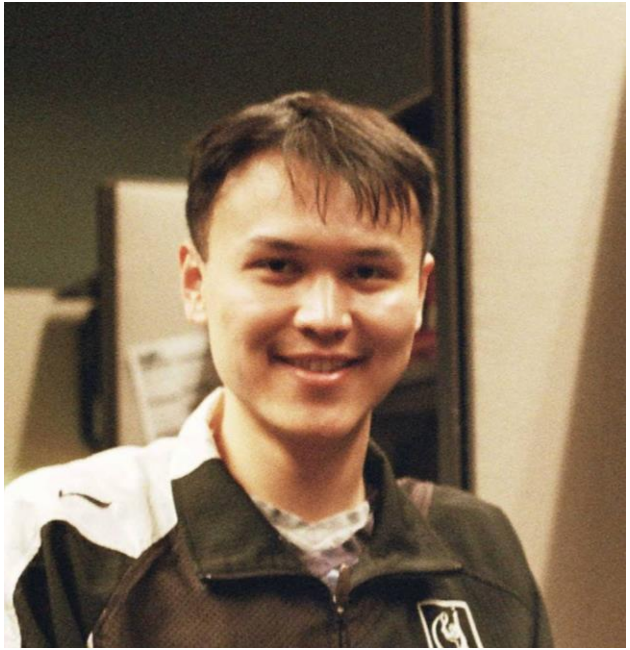
Course Website: https://harviu.github.io/CIS371/

Subject to change throughout the semester

Special Thanks to Dr. Yong Zhuang for construction the original version of this website

About Me

- Haoyu Li, Assistant Professor joined GVSU in 2024
- PhD in Computer Science @ The Ohio State University
- BS in Psychology
- Research Interest: Visualization and Computer Graphics
- Personal Interests:
    - Badminton, board games, and hiking



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